

Largest induced trees, girth, and distance from the periphery: a proof of Graffiti.pc's Conjecture 142

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Abstract

For a finite simple connected graph G , let $t(G)$ be the largest order of an induced subgraph that is a tree, let $g(G)$ be the girth, let $\text{Per}(G)$ be the set of peripheral vertices, and put $f(G) = \max_{x \in V(G)} d(x, \text{Per}(G))$. We prove

$$t(G) \geq \frac{2}{3}g(G) + f(G),$$

which is Conjecture 142 of the conjecture-making program Graffiti.pc (DeLaViña, *Written on the Wall II*, 2005). In the cyclic case we prove the stronger integral form $t(G) \geq f(G) + \lceil 2g(G)/3 \rceil$. The main ingredient is a rooted shortest-cycle lemma: in a graph of girth at least five, any three prescribed vertices can be placed in a tree obtained from a shortest cycle by deleting one cycle vertex and attaching an admissible induced forest. A three-point metric estimate then reduces the result to exact arithmetic, with one extremal family handled separately.

1 Introduction

Graffiti.pc, developed by DeLaViña [1, 2] as a successor of Fajtlowicz's Graffiti program [5], generates conjectural inequalities among graph invariants. Its conjectures are collected in *Written on the Wall II* [3]; a curated selection is maintained by West [4]. Several consecutive entries concern the maximum order of an induced tree, an invariant systematically studied by Erdős, Saks, and Sós [6].

Let G be a finite simple connected graph. The *periphery* $\text{Per}(G)$ is the set of vertices of eccentricity $\text{diam}(G)$, and the set-eccentricity of the periphery is

$$f(G) := \max_{x \in V(G)} d(x, \text{Per}(G)).$$

Conjecture 142 of *Written on the Wall II* asserts

$$t(G) \geq \frac{2}{3}g(G) + f(G).$$

Here and in the formal statement, the girth of an acyclic graph is taken to be zero. We prove the following stronger cyclic statement.

Theorem 1.1 (Graffiti.pc Conjecture 142). *Let G be a finite simple connected graph containing a cycle. If $g = g(G)$ and $f = f(G)$, then*

$$t(G) \geq f + \left\lceil \frac{2g}{3} \right\rceil.$$

Consequently every finite simple connected graph satisfies $t(G) \geq \frac{2}{3}g(G) + f(G)$.

The proof is self-contained apart from elementary facts about finite graphs. Its structural core is Lemma 3.1, which turns three metric terminals into an admissible forest attached to a shortest cycle. The resulting metric inequality, Lemma 4.1, is then combined with a short classification of the equality case in which the admissible forest has only one vertex.

2 Preliminaries

Write $D = \text{diam}(G)$ and $B = \text{Per}(G)$. A shortest path is induced, so

$$t(G) \geq D + 1. \quad (2.1)$$

A shortest cycle K is chordless and isometric: for $u, v \in V(K)$, their graph distance equals their shorter arc-distance in K . Deleting one vertex of K therefore gives

$$t(G) \geq g - 1. \quad (2.2)$$

If $f \geq 1$, then

$$D \geq f + 1. \quad (2.3)$$

Indeed, always $f \leq D$; equality would give a vertex x at distance D from every peripheral vertex, hence $\text{ecc}(x) = D$ and $x \in B$, contradicting $f = d(x, B) \geq 1$.

For a shortest cycle K , define $M(K)$ to be the maximum of $|F|$ over all sets $F \subseteq V(G) \setminus V(K)$ for which $G[F]$ is a forest and there is a vertex $z \in V(K)$ such that every component C of $G[F]$ sends exactly one edge into $V(K) \setminus \{z\}$. Edges from C to z are unrestricted.

Lemma 2.1 (Cycle-forest extension). *For every shortest cycle K ,*

$$t(G) \geq g - 1 + M(K).$$

Proof. Choose F, z realizing $M(K)$ and let c be the number of components of $G[F]$. The graph induced by $(V(K) \setminus \{z\}) \cup F$ is connected. Its edges consist of the $g - 2$ edges of the path $K - z$, the $|F| - c$ forest edges, and one attachment edge for each of the c components. Thus it has $g - 1 + |F|$ vertices and one fewer edge, and hence is a tree. $\square$

Lemma 2.2 (An extra cycle vertex). *Suppose $f \geq 1$. If $g \geq 4$, then $t(G) \geq g$; if $g \geq 5$, then $M(K) \geq 1$ for every shortest cycle K .*

Proof. If $V(G) = V(K)$, chordlessness and connectedness give $G = C_g$, whose every vertex is peripheral, contrary to $f \geq 1$. Hence some vertex $y \notin K$ is adjacent to K . When $g \geq 5$, it has exactly one neighbour on K : two neighbours and the shorter arc between them would form a cycle of length at most $\lfloor g/2 \rfloor + 2 < g$. Taking $F = \{y\}$ and deleting a cycle vertex other than its root proves both assertions. For $g = 4$, the only additional possibility is that y has two antipodal neighbours on K ; deleting one of them again leaves an induced tree on four vertices. $\square$

We shall also need two elementary connector lemmas.

Lemma 2.3 (Three terminals). *Every three distinct vertices of a connected triangle-free graph lie in a common induced tree.*

Proof. Let P be an a - b geodesic and let $R = r_0 \cdots r_\ell$ be a shortest path from $c = r_0$ to P , where $c \notin P$. Put $q = r_{\ell-1}$ and $R^\circ = R - r_\ell$. Only q can have additional neighbours on P . Any two such neighbours have P -distance at most two, and triangle-freeness excludes distance one; hence q has at most two neighbours on P . With one neighbour, $P \cup R^\circ$ induces a tree. With two neighbours u, v , their P -subpath is umv ; deleting m joins the two components of $P - m$ through q and yields a connected induced graph with one fewer edge than vertices. Since $m \notin \{a, b\}$, this is the required tree. $\square$

Lemma 2.4 (Cycles in a minimal connector). *Let J be connected and $Q \subseteq V(J)$. If no proper induced connected subgraph of J contains Q , then every cycle L of J has $|V(L)| \leq |Q|$.*

Proof. For every $q \in Q$, choose a shortest path from q to L , and denote its unique first vertex on L by $c(q)$. Fix $v \in L$. The path $L - v$ lies in one component J_v of $J - v$. Minimality implies that some $q_v \in Q$ is outside J_v . Its chosen path to L must therefore end at v , so $c(q_v) = v$. Distinct vertices of L give distinct terminals, which injects $V(L)$ into Q . $\square$

3 A rooted shortest-cycle lemma

Lemma 3.1 (Rooted shortest cycle). *Assume $g \geq 5$, and let $S \subseteq V(G)$ with $|S| \leq 3$. There exist a shortest cycle K , a vertex $z \in V(K) \setminus S$, and a set $F \subseteq V(G) \setminus V(K)$ such that*

$$S \subseteq (V(K) \setminus \{z\}) \cup F,$$

$G[F]$ is a forest, and every component C of $G[F]$ satisfies

$$e(C, V(K) \setminus \{z\}) = 1.$$

Proof. The empty case is immediate. Fix an arbitrary shortest cycle K , and choose $X \supseteq V(K) \cup S$ of minimum cardinality such that $H := G[X]$ is connected.

Every component C of $H - V(K)$ contains a terminal. Indeed, it has an edge to K , since H is connected; if $S_C := S \cap V(C)$ were empty, the whole component could be deleted while preserving connectedness and all required vertices. Put

$$A_C := \{a \in V(C) : N_G(a) \cap V(K) \neq \emptyset\}.$$

Form J_C from $G[C]$ by adjoining a new vertex r_C adjacent precisely to A_C . This graph is vertex-minimal among induced connected subgraphs containing $S_C \cup \{r_C\}$. Otherwise, replacing C by a smaller connector would leave every resulting component attached to K and would contradict the choice of X .

Lemma 2.4 shows that every cycle in J_C has length at most $|S_C| + 1 \leq 4$. A cycle contained in $G[C]$ would also be a cycle of G and would have length at least $g \geq 5$. Consequently

$$G[C] \text{ is a tree.} \tag{3.1}$$

Let R_C be the minimal subtree of C containing A_C . We claim

$$|V(R_C)| \leq |S_C|. \tag{3.2}$$

For $v \in V(R_C)$, let Q_v be the component of $J_C - v$ containing r_C . Minimality supplies $s_v \in S_C \setminus V(Q_v)$. For $s \in S_C$, let $c(s)$ be the first vertex of R_C on the unique path from s to R_C . Every nonempty component of $R_C - v$ contains a vertex of A_C , by the minimality of R_C . Thus, if $v \neq c(s)$, the path from s to $c(s)$ and then to such an attachment gives an s - r_C path in $J_C - v$; hence $s \in Q_v$. Applying this to s_v gives $c(s_v) = v$. The map $v \mapsto s_v$ is injective, proving (3.2).

Every $a \in A_C$ has exactly one neighbour on K . Existence follows from the definition, while two distinct neighbours and their shorter K -arc would form a cycle of length at most $\lfloor g/2 \rfloor + 2 < g$. Denote the unique root by $\rho(a)$. If $a, a' \in A_C$ are distinct, then

$$d_C(a, a') \leq |V(R_C)| - 1 \leq 2 \quad (3.3)$$

and their roots are distinct: equal roots, together with the a - a' path in C , would form a cycle of length at most four. Moreover, the two root edges, that path, and a shorter arc of K give

$$d_K(\rho(a), \rho(a')) + d_C(a, a') + 2 \geq g. \quad (3.4)$$

These inequalities sharply restrict the number of attachments. If $g \geq 9$, then two attachments would make the left side of (3.4) at most $\lfloor g/2 \rfloor + 4 < g$, so there is only one. For $g \in \{7, 8\}$, two attachments cannot be adjacent in C ; three attachments would fill the three vertices of R_C and contain an adjacent pair. For $g = 6$, if $R_C = p - q - r$ consisted of three attachments, applying (3.4) to p, q and to q, r would force both $\rho(p)$ and $\rho(r)$ to be the unique antipode of $\rho(q)$ on the 6-cycle, contrary to distinctness. Thus every component has at most two attachments, except that for $g = 5$ one component may have three. Also, at most one component has two or more attachments, because such a component contains at least two terminals and $|S| \leq 3$.

Suppose first that every component has one attachment a_C . The set

$$Q = (S \cap V(K)) \cup \{\rho(a_C) : C \text{ a component of } H - V(K)\}$$

has cardinality at most $|S| \leq 3$. Since $g \geq 5$, choose $z \in V(K) \setminus Q$ and take $F = X \setminus V(K)$. Equation (3.1) gives a forest, each component has its single attachment in Q , and $z \notin S$.

Next suppose that one component C_0 has exactly two attachments with roots u, v . It contains at least two terminals, so the set

$$Q = (S \cap V(K)) \cup \{\rho(a_C) : C \neq C_0\}$$

has size at most one. Choose $z \in \{u, v\} \setminus Q$ and again put $F = X \setminus V(K)$. Every ordinary component has one edge into $K - z$, while C_0 has exactly the two root edges and loses exactly one of them. Again $z \notin S$.

It remains to handle three attachments. Necessarily $g = 5$, $|V(R_C)| = |S_C| = |S| = 3$, there are no other off-cycle components, and $R_C = p - q - r$ with all three vertices attachments. Write

$$u = \rho(p), \quad v = \rho(q), \quad w = \rho(r).$$

Equation (3.4) gives $d_K(u, v) = d_K(v, w) = 2$. In cyclic order, write $K = u\alpha v\beta wu$. Then

$$K' = u\alpha vqpu$$

is another shortest 5-cycle.

The injection used to prove (3.2) is now a bijection. Hence there are distinct terminals s_p, s_q, s_r with $c(s_t) = t$. Let L_t be the unique t - s_t path in the tree C , and set

$$F' = (V(L_p) \setminus \{p\}) \cup (V(L_q) \setminus \{q\}) \cup V(L_r), \quad z = \alpha.$$

The three displayed path-pieces are pairwise disjoint and anticomplete, for an edge between two of them would create a cycle in the tree C . The first two nonempty pieces attach once to K' at p and q , respectively. The third attaches once through the edge rq ; its other old-cycle edge rw does not meet K' . Thus F' is admissible for (K', z) . Finally each s_t belongs either to its path-piece or, when $s_p = p$ or $s_q = q$, directly to $K' - z$. This completes the exceptional case and the proof. $\square$

4 The three-point bridge

Let

$$\mu := \max\{M(K) : K \text{ is a shortest cycle of } G\}.$$

Lemma 4.1 (Three-point inequality). *If $g \geq 5$, then every three distinct vertices a, b, c satisfy*

$$d(a, b) + d(b, c) + d(c, a) \leq g + 2\mu. \quad (4.1)$$

Proof. Apply Lemma 3.1 to $S = \{a, b, c\}$ and obtain K, z, F . Retain all edges of K , all edges inside the components of $G[F]$, and each component's unique edge into $K - z$; omit possible edges to z . The resulting spanning subgraph U of $G[K \cup F]$ is a single cycle K with rooted trees attached. It has exactly $|F|$ edges outside K : a component with k vertices contributes $k - 1$ internal edges and one root edge.

For each pair of terminals, use their unique paths to their roots and a shortest arc of K between the roots. An off-cycle edge separates from its root q of the three terminals. If its rooted tree contains h terminals, then its total multiplicity in the three selected paths is

$$q(h - q) + q(3 - h) = q(3 - q) \leq 2.$$

Thus the total off-cycle contribution is at most $2|F|$.

The sum of the three pairwise distances between any three roots on a cycle of length g is at most g . Indeed, if the roots are distinct and the three intervening arc-lengths are all at most $g/2$, their sum is exactly g ; if one arc exceeds $g/2$, the other two provide the shorter route for its endpoints and the sum is smaller. Coincident roots are immediate. Hence the three chosen paths have total length at most $g + 2|F| \leq g + 2M(K) \leq g + 2\mu$. Graph distances can only be shorter. $\square$

Corollary 4.2. *Assume $g \geq 5$ and $f \geq 1$. Then*

$$2\mu + g \geq 3f + 1. \quad (4.2)$$

Proof. Choose x with $d(x, B) = f$ and a diametral pair b, w . Both b, w are peripheral, so $d(x, b), d(x, w) \geq f$ and $d(b, w) = D \geq f + 1$ by (2.3). Apply Lemma 4.1 to x, b, w . $\square$

The estimate is almost sufficient by itself. The only delicate integer case is disposed of by the following sharp classification.

Lemma 4.3 (The case $\mu = 1$). *Assume $g \geq 5$ and $f \geq 1$. If $\mu = 1$, then G is obtained from C_g by attaching one pendant vertex to one cycle vertex. Consequently*

$$f \leq \left\lfloor \frac{g+2}{4} \right\rfloor. \quad (4.3)$$

Proof. Fix a shortest cycle K . Since $M(K) \leq \mu = 1$, no vertex can have distance two from K : the first two off-cycle vertices on a geodesic to K would form an admissible two-vertex path. Thus every vertex outside K is adjacent to K , and by $g \geq 5$ it has a unique root.

There cannot be two such vertices u, v . If they are nonadjacent, choose z different from both roots; the two singleton components are admissible. If they are adjacent, their roots are distinct (otherwise there is a triangle), and choosing z to be one root makes $\{u, v\}$ a connected admissible component. Either way $M(K) \geq 2$, a contradiction. Since $f \geq 1$, there is exactly one outside vertex, and G is the stated tadpole.

Put $k = \lfloor g/2 \rfloor$, let y be the pendant vertex, and let v_0 be its root. The diameter is $k + 1$; the periphery consists of y and the cycle vertices at cycle-distance k from v_0 . Every cycle vertex at position i on one of the v_0 -antipode arcs is at distance at most

$$\min(i + 1, k - i) \leq \left\lfloor \frac{k + 1}{2} \right\rfloor = \left\lfloor \frac{g + 2}{4} \right\rfloor$$

from the displayed peripheral vertices. The pendant vertex itself is peripheral, proving (4.3). $\square$

5 Completion of the proof

Proposition 5.1. *Theorem 1.1 holds when $g \geq 5$ and $f \geq 1$.*

Proof. Put $a = \lfloor g/3 \rfloor$. Since $\lceil 2g/3 \rceil = g - a$, the target is

$$t(G) \geq f + g - a. \quad (5.1)$$

If $f \leq a - 1$, the cycle bound (2.2) gives $t(G) \geq g - 1 \geq f + g - a$. Hence write $f = a + s$ with $s \geq 0$.

Lemma 2.2 gives $\mu \geq 1$, while Lemma 2.1 gives $t(G) \geq g - 1 + \mu$. It therefore suffices to prove

$$\mu \geq s + 1. \quad (5.2)$$

Corollary 4.2 and the three residues of g give:

$$\begin{array}{ll} g = 3a : & 2\mu \geq 3s + 1, \\ g = 3a + 1 : & 2\mu \geq 3s, \\ g = 3a + 2 : & 2\mu \geq 3s - 1. \end{array}$$

In the first case, $\mu \leq s$ would imply $2\mu \leq 2s < 3s + 1$. In the second, $s = 0$ is covered by $\mu \geq 1$, while for $s \geq 1$ we have $2s < 3s$. In the third, $s = 0$ is again immediate, and $s \geq 2$ gives $2s < 3s - 1$.

The only remaining possibility is $g = 3a + 2$ and $s = 1$. If $\mu = 1$, Lemma 4.3 yields

$$f \leq \left\lfloor \frac{3a + 4}{4} \right\rfloor \leq a,$$

because $a \geq 1$, contradicting $f = a + 1$. Hence $\mu \geq 2 = s + 1$. This proves (5.2), and therefore (5.1). $\square$

Proposition 5.2. *Theorem 1.1 holds for $g = 3$ and $g = 4$ when $f \geq 1$.*

Proof. For $g = 3$, equations (2.1) and (2.3) give $t(G) \geq D + 1 \geq f + 2 = f + \lceil 2g/3 \rceil$.

Let $g = 4$. Choose an f -realizer x and a diametral pair b, w . The graph is triangle-free, so Lemma 2.3 supplies an induced tree T containing x, b, w . Let T_0 be the minimal subtree containing them. Every edge of T_0 lies on exactly two of the three terminal-to-terminal paths; therefore

$$2|E(T_0)| = d_T(x, b) + d_T(x, w) + d_T(b, w) \geq 2f + D \geq 3f + 1.$$

If $f \geq 2$, then

$$|V(T)| \geq 1 + \left\lceil \frac{3f+1}{2} \right\rceil \geq f + 3.$$

If $f = 1$, Lemma 2.2 gives $t(G) \geq g = 4 = f + 3$. $\square$

Proof of Theorem 1.1. The preceding two propositions cover cyclic graphs with $f \geq 1$. If G is cyclic and $f = 0$, then $g - 1 \geq \lceil 2g/3 \rceil$ for $g \geq 3$, so (2.2) applies. This proves the integral cyclic assertion.

If G is acyclic, connectedness makes G itself a tree, so $t(G) = |V(G)|$ and $f \leq D \leq |V(G)| - 1$ (with the one-vertex case immediate). Since $g(G) = 0$, the real-valued Formal Conjectures inequality also follows. Finally, $\lceil 2g/3 \rceil \geq 2g/3$, completing the proof. $\square$

6 Verification and formalization status

The proof above was checked in two independent forms. First, a separate constructive proof based on descents to an arbitrary shortest cycle was implemented literally: every branch produces its cycle vertex z , its admissible forest, or its induced path, and a fresh checker verifies inducedness, connectedness, acyclicity, edge multiplicities, and the claimed cardinality. On a corpus of 10,776 connected cyclic graphs (the complete NetworkX atlas through seven vertices, structured extremal families, and seeded random and adversarial graphs through 32 vertices), the validator reported no failure. Repeating every free choice with two independent random seeds again gave no failure, for 32,328 checked certificates in all. These computations are supporting checks, not a substitute for the proof.

The exact statement of Conjecture 142 is formalized in the Google DeepMind *Formal Conjectures* repository [8, 9], using the repository’s largest-induced-tree invariant, Mathlib’s natural-valued girth, and the set-eccentricity of the boundary (peripheral) vertices. A compiled Lean 4 skeleton already closes the acyclic and $f = 0$ branches, and the reusable cycle–forest extension API has been checked. At the date of this draft, the rooted shortest-cycle lemma and the final cyclic branch have not yet been fully formalized; we therefore make no claim here of a completed Lean proof.

Remark 6.1 (Sharpness). Equality occurs throughout the $g = 3$ diameter branch. Among the exhaustively checked small graphs, the only equality class of girth greater than three is the 6-cycle with one pendant vertex: here $f = 2$ and $t(G) = 6 = f + \lceil 2g/3 \rceil$. In the proof this is exactly the $\mu = 1$ tadpole case of Lemma 4.3.

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